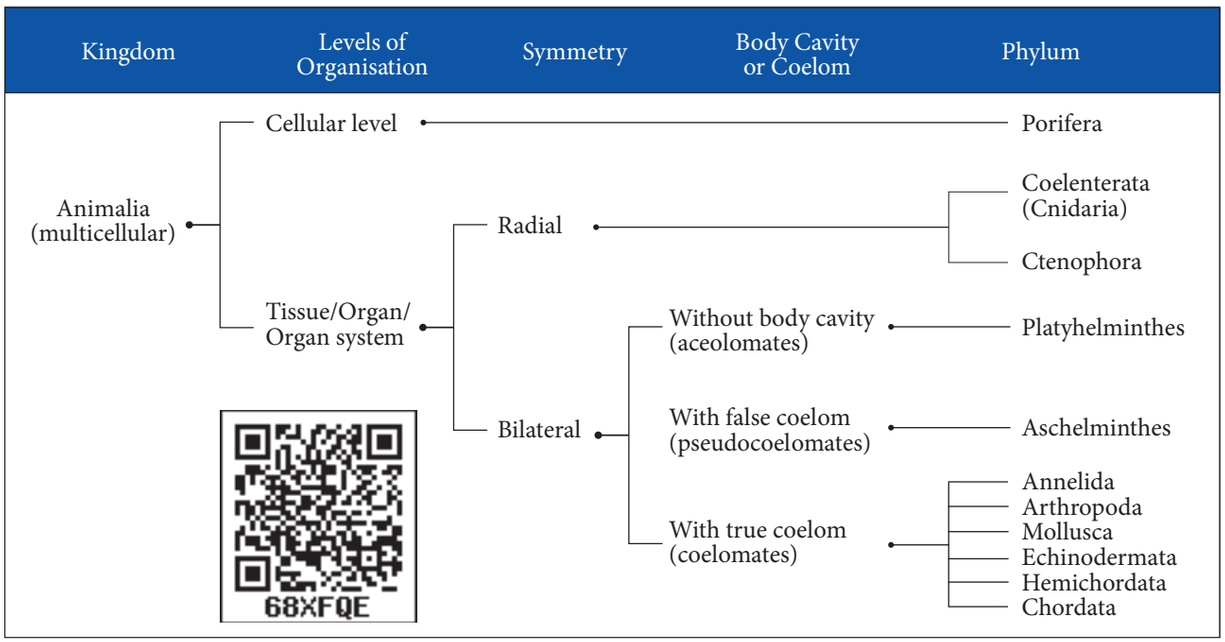


Figure 2.8 Classification of Kingdom Animalia based on common fundamental features

Grade 1: Radiata

Among the eumetazoa, a few animals have an organisation of two layers of cells, the outer ectoderm and inner endoderm, separated by a jelly like mesoglea. They are radially symmetrical and are diploblastic. Examples: Cnidarians (sea anemone, jelly fish) and Ctenophores (comb jellies).

Grade 2: Bilateria

The eumetazoans other than Radiata, show organ level of organisation and are bilaterally symmetrical and triploblastic. The grade Bilateria includes two taxonomic levels called **Division.**

Division 1: Protostomia

(*Proto*: first; *stomium*: mouth)

Protostomia includes the eumetazoans in which the embryonic blastopore develops into mouth. This division includes three subdivisions namely **acoelomata, pseudocoelomata and schizocoelomata.**

Division 2: Deuterostomia

(*deuteron*: secondary; *stomium*: mouth)

Eumetazoans in which anus is formed from or near the blastopore and the mouth is formed away from the blastopore. It includes only one subdivision **Enterocoelomata.** They have a true coelom called enterocoel, formed from the archenteron.

2.3 Non Chordates (Invertebrata)

2.3.1 Phylum: Porifera

(L. *poros*-pore; *ferre*-to bear)

These pore bearing animals are commonly called **sponges.** They are aquatic, mostly marine, asymmetrical and a few species live in freshwaters. They are primitive, multicellular, sessile animals with cellular level of organisation in which the cells are loosely arranged. They are either radially symmetrical or asymmetrical animals.

They possess a water transport system or **canal system** where water enters through minute pores called **ostia** lining the body wall through which the water enters into a

TN_GOVT_XI_Zoology_Ch02.indd 19 09-01-2020 5.11.59 PM